

ACCEPTANCE SAMPLING

Driving Supply Chain Quality in Real-World Industries

Course: Quality Management in Supply Chain Systems

Program: Master of Supply Chain Management | KFUPM Business School

Reference: Foster (2017), Chapter 9A | ISO 2859-1 | MIL-STD-1916



Opening Case: SABIC Polymer Resin Procurement

Real-world context • Saudi Arabia

The Challenge

- SABIC sources high-density polyethylene (HDPE) pellets from 12+ global suppliers.
- Each shipment: 20–50 tonnes. Full 100% inspection is impossible and cost-prohibitive.
- Contaminated batches cause defective final products, customer complaints & recall costs.
- SABIC uses **MIL-STD-105E (now ISO 2859-1)** Acceptance Sampling to inspect incoming resin lots.
- Sampling plan: $n = 80$, $AQL = 1.5\%$ → random sample drawn; lot accepted or rejected based on c .

Key Numbers

Lot Size (N)	2,000 units
Sample (n)	80 units
AQL	1.5%
Accept if defects $\leq$	$c = 3$

Outcome

- Rejected 3 supplier lots in Q1 2023 alone.
- Saved est. \$2.4M in downstream rework.
- Supplier corrective action initiated.

Learning Objectives

By the end of this session, you will be able to...

1

Explain when and why acceptance sampling is necessary in supply chains

2

Define AQL, LTPD, producer's risk (α), and consumer's risk (β)

3

Interpret and construct OC (Operating Characteristic) curves

4

Design single- and double-sampling plans (n and c)

5

Apply sampling to real Saudi/global industrial contexts

6

Run a classroom simulation and make accept/reject decisions



OPEN DISCUSSION

Think Before We Dive In...

You are a Quality Manager at a Saudi petrochemical company.
A supplier delivers 10,000 valves per shipment. Testing every valve destroys it.
You must decide: how many to test, and what level of defects is 'acceptable'?

Q1: What % defective is acceptable before you reject the whole shipment?

Q2: Who bears the cost if you accept a bad lot? If you reject a good lot?

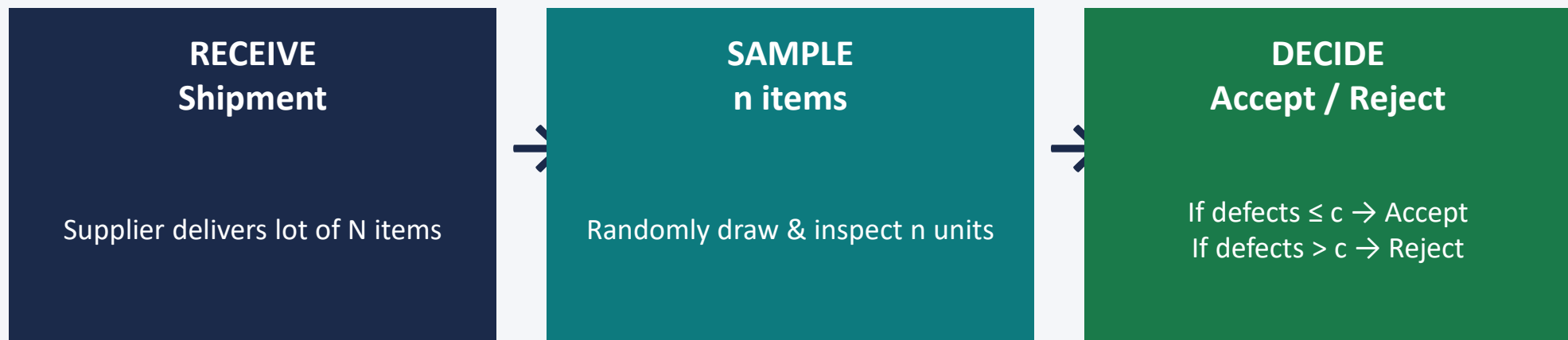
Q3: Could you use statistics to make this decision more scientific?

 5 Minutes — Discuss with a partner

What is Acceptance Sampling?

Definition & Scope

A statistical technique to decide whether to accept or reject a batch (lot) of goods by inspecting a sample — without testing every item.



When is it Used?

- Destructive testing (e.g., tensile tests, pressure valves, pharmaceutical potency)
- Very large lot sizes where 100% inspection is impractical or too expensive
- New or unproven suppliers — pharmaceutical ingredients, electronic components

Core Concepts: AQL, LTPD, α & β

The four pillars of any sampling plan

AQL

Acceptable Quality Level

Max % defective that is still considered satisfactory. Linked to Producer's Risk (α).

e.g., AQL = 1.5% $\rightarrow$ up to 15 defects per 1,000 units is "acceptable"

LTPD

Lot Tolerance Percent Defective

Quality level that is barely tolerable — anything worse should be rejected. Linked to Consumer's Risk (β).

e.g., LTPD = 8% $\rightarrow$ lots with 8%+ defects should almost always be rejected

α

Producer's Risk (Type I Error)

Probability of rejecting a GOOD lot. The supplier suffers — their high-quality lot is sent back.

Typical $\alpha = 0.05$ (5%)

β

Consumer's Risk (Type II Error)

Probability of accepting a BAD lot. The buyer suffers — poor quality enters production.

Typical $\beta = 0.10$ (10%)

Risk Matrix: Producer vs. Consumer

Type I & Type II Errors in Quality Decisions

DECISION → REALITY ↓	Accept Lot ✓	Reject Lot ✗
Lot is GOOD (Low defects)	✓ Correct Decision (No error)	✗ Type I Error Producer's Risk (α) Good lot rejected!
Lot is BAD (High defects)	✗ Type II Error Consumer's Risk (β) Bad lot accepted!	✓ Correct Decision (No error)

Goal: Minimize both α AND β through careful design of n and c

Designing a Sampling Plan: n and c

The two numbers that define every single-sampling plan

n = Sample size | **c** = Maximum number of defectives allowed before REJECTING the lot

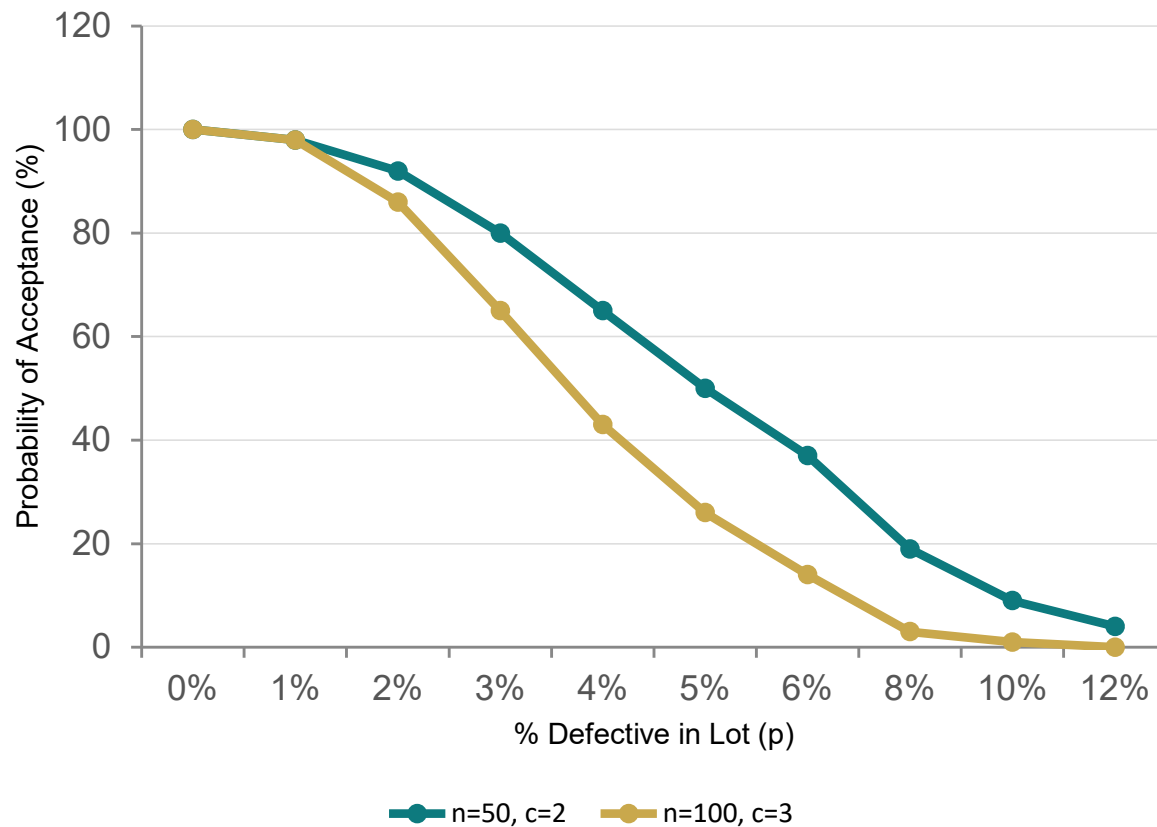
Worked Example

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Lot of $N = 2,000$ valves arrives from supplier	Draw random sample: $n = 50$ valves	Inspect each valve for defects	Count defectives found: d	If $d \leq c$ (e.g., $c = 2$): ACCEPT If $d > 2$: REJECT entire lot

💡 Management sets AQL, LTPD, α and β — then n and c are calculated to satisfy those targets simultaneously.

Operating Characteristic (OC) Curve

The visual backbone of acceptance sampling



AQL Region

Low % defective → High probability of acceptance (correct!)

LTPD Region

High % defective → Low probability of acceptance (correct!)

Reading the Chart

- X-axis: True % defective in lot
- Y-axis: $P(\text{Accept})$ the lot
- Steeper curve = better discrimination
- Larger n → steeper, more precise curve

Building an OC Curve — Step by Step

Example: $n = 50$, $c = 2$ using Poisson approximation

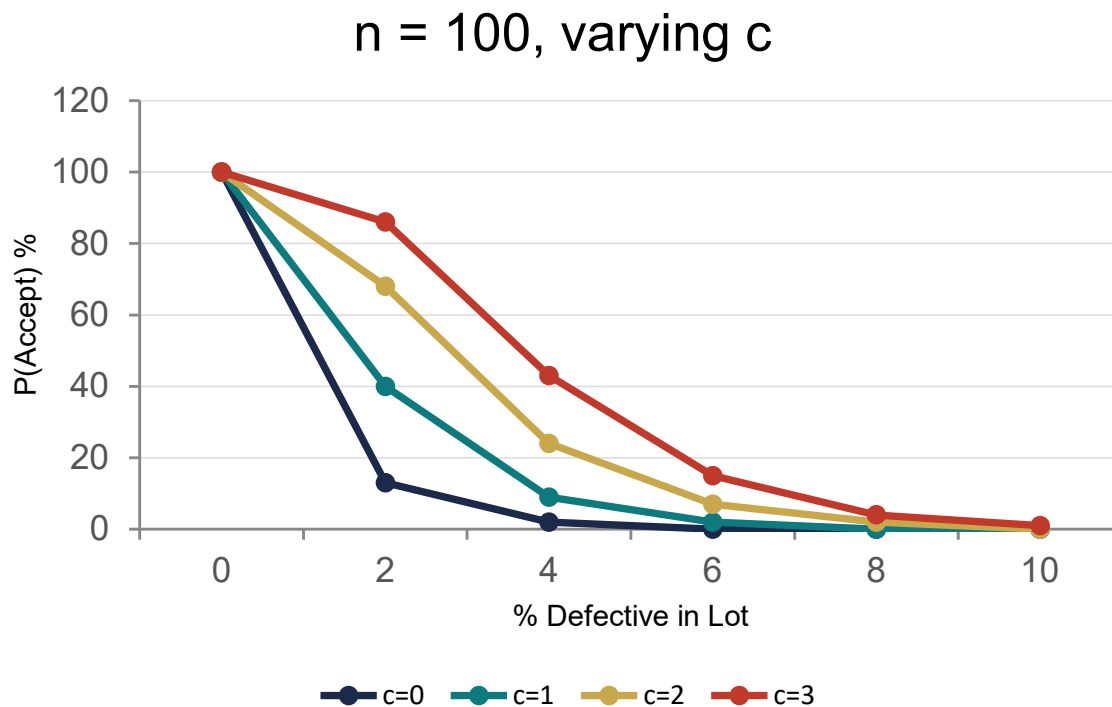
Method: Use Poisson distribution — $P(\text{accept}) = P(X \leq c \mid np)$ where np = mean number of defectives in sample

p (% def)	np = 50×p	$P(X \leq 2 \mid np)$	Decision
0%	0.00	1.000	Accept
1%	0.50	0.986	Accept
2%	1.00	0.920	Accept
3%	1.50	0.809	Accept
4%	2.00	0.677	Accept
5%	2.50	0.544	Accept
6%	3.00	0.423	Accept/Reject zone
8%	4.00	0.238	Likely Reject
10%	5.00	0.125	Reject
12%	6.00	0.062	Reject

Source: Poisson CDF tables / Foster (2017) Table 9A-2

Effect of n and c on the OC Curve

How sample size and acceptance number shape your plan



↑ Increase c

Easier to accept lots → raises consumer's risk (β)
↑
More lenient on supplier

↓ Decrease c

Harder to accept lots → raises producer's risk (α)
↑
More stringent inspection

↑ Increase n

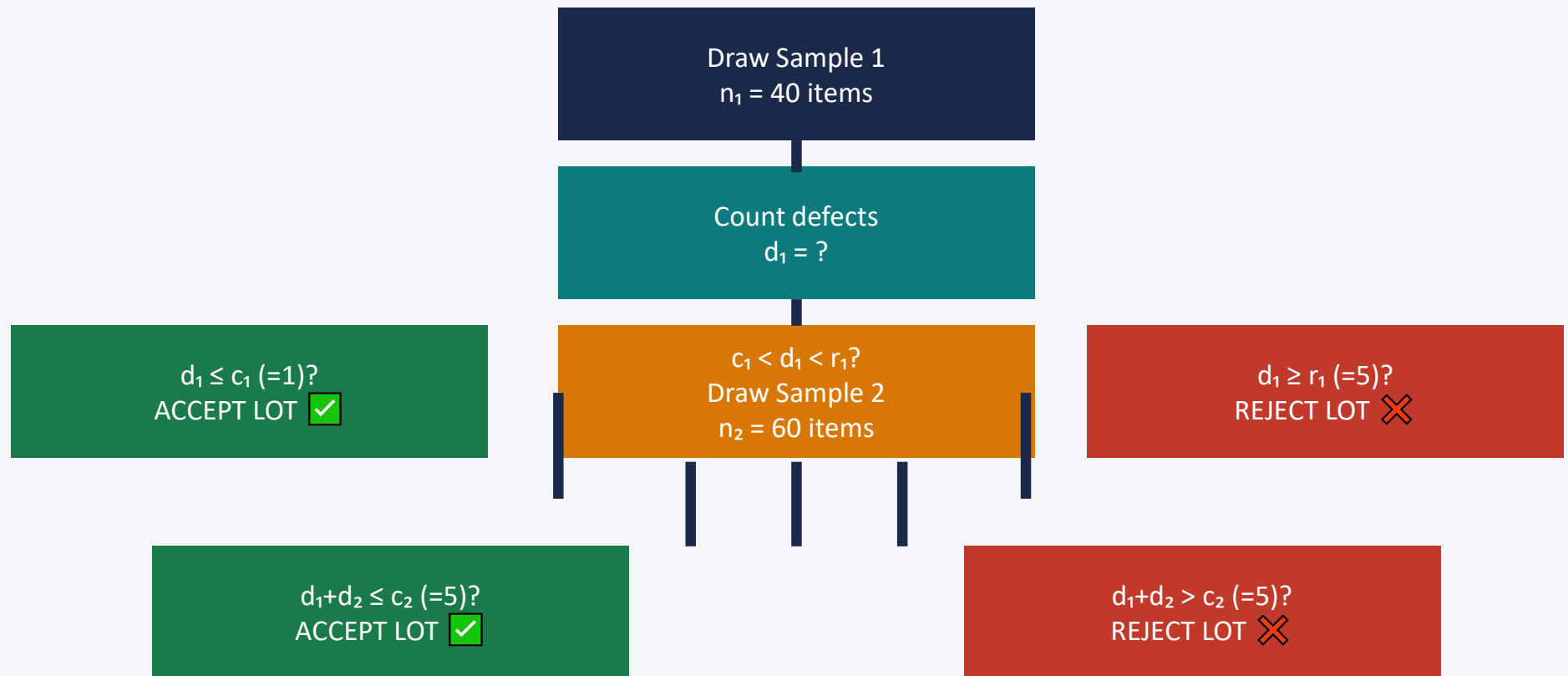
OC curve gets steeper → better discrimination between good and bad lots

Balance n & c

The art is finding n and c that satisfy both $\alpha \leq 0.05$ and $\beta \leq 0.10$ simultaneously

Double Sampling Plans

Two-stage decision process for greater efficiency



✓ Advantage: On average, double sampling inspects FEWER items than single sampling for the same quality protection (saves cost)

Real Case Study: SFDA & Pharmaceutical Imports

Saudi Food & Drug Authority — ISO 2859-1 in Action

Case Background

The Saudi Food & Drug Authority (SFDA) regulates the quality of imported pharmaceutical products including tablets, vials, and medical devices.

SFDA inspectors apply ISO 2859-1 Level II inspection at port of entry (King Abdulaziz Port, Dammam).

A shipment of 10,000 antibiotic tablets from a European supplier was sampled: $n = 200$, AQL = 0.65%.

Upon inspection: 3 defective tablets found (cracked coating, incorrect label). Since $3 > c = 1$ (for AQL 0.65%), the ENTIRE lot was rejected.

The supplier had to re-manufacture and re-ship at full cost. SFDA issued a corrective action request.

Sampling Plan Parameters

Lot Size (N)	10,000 tablets
Sample Size (n)	200 tablets
AQL	0.65%
Acceptance No. (c)	1
Defects Found	3 → REJECT

Outcome & Lessons

- Lot rejected → supplier financial loss
- SFDA protected Saudi patients from substandard drugs
- Demonstrates sampling's role as a supply chain 'gatekeeper'
- AQL 0.65% = stringent for pharmaceuticals (life-critical)

Global Case Study: Toyota's Supplier Quality System

Acceptance Sampling + Jidoka at Scale

Toyota Motor Corporation uses acceptance sampling as part of its Global Production System (TPS) for incoming auto parts — but pairs it with supplier development to reduce the need for it over time.

The Challenge

- Toyota sources 30,000+ parts from 500+ global suppliers
- Some components are safety-critical (brakes, airbags)
- Full 100% inspection of every part is impossible in real-time production

Their Approach

- New suppliers: use tightened inspection (ISO 2859-1 Level III, AQL = 0.4%)
- Mature certified suppliers: reduced inspection (Level I, AQL = 1.0%)
- Critical safety parts: 100% AQL = 0% with statistical SPC monitoring

Results

- Defect rate dropped from 1,200 ppm (1990) to <5 ppm (2020)
- Supplier quality certification reduces inspection cost by ~40%
- Recall costs reduced by billions through early lot rejection

Industry Standards for Acceptance Sampling

The global frameworks practitioners use

ISO 2859-1	Sampling procedures for inspection by attributes Most widely used globally. Defines inspection levels, AQL tables, tightened/normal/reduced rules. Used by SFDA, SABIC, automotive.
MIL-STD-1916	US DoD Sampling Procedures Replaced MIL-STD-105E. Zero-based acceptance sampling. Used in defense procurement and aerospace supply chains.
ANSI/ASQ Z1.4	US version of ISO 2859-1 Used extensively in US manufacturing. Nearly identical to ISO 2859-1 but with US customary units and conventions.
ISO 3951-1	Sampling by variables Uses actual measurements (not just pass/fail). More efficient — needs smaller sample for same confidence. Used in pharmaceuticals, chemicals.
ISO 2859-2	Sampling for isolated lots For single one-time shipments (not ongoing supply). Common in construction materials, capital equipment.



CLASSROOM SIMULATION

You Are the Quality Inspector

Scenario: Saudi Dates Export — Madinah Packaging Co.

A lot of $N = 500$ date packages (tamr) is ready for export. Each group represents the Quality Control team. Your sampling plan: $n = 20$, $c = 2$.

STEP 1 2 min

Each group draws 20 cards from a shuffled deck (face cards = DEFECT; others = OK)

STEP 2 2 min

Count defective cards in your sample. Record d (number of defects).

STEP 3 1 min

Apply rule: if $d \leq 2 \rightarrow$ Accept lot. If $d > 2 \rightarrow$ Reject lot. Report decision.

STEP 4 5 min

Class discussion: Who rejected? Who accepted? Why? What does this tell us about sampling risk?

Simulation Debrief & Results

What does your sample tell you about the lot?

Group Results Table

Group	d (defects found)	Decision
Group A	—	<i>Accept / Reject</i>
Group B	—	<i>Accept / Reject</i>
Group C	—	<i>Accept / Reject</i>
Group D	—	<i>Accept / Reject</i>
Group E	—	<i>Accept / Reject</i>
Group F	—	<i>Accept / Reject</i>

Key Takeaway: Sampling variability is real. A good plan minimizes both errors — but never eliminates them.

Discussion Questions

Q1: Did all groups make the same decision? Why not?

Q2: If the true defect rate is 5%, what is $P(\text{accept})$ for $n=20$, $c=2$?

Q3: A group with $d=2$ accepted — was that the 'right' call?

Q4: How would your decision change if $c=1$ instead of $c=2$?

Q5: What is the cost implication of a wrong decision here?

Acceptance Sampling Across Saudi Industries

Where Vision 2030 meets quality science



Petrochemicals (SABIC, ARLANXEO)

Incoming polymer resin, catalyst, packaging — ISO 2859-1 Level II, AQL 0.65–1.5%



Oil & Gas (Saudi Aramco)

Inspection of pipes, valves, flanges from global suppliers — API Q1 + acceptance sampling clauses



Pharmaceuticals (SFDA, Tabuk Pharma)

Import inspection of APIs and finished drugs — ICH Q6A + ISO 2859-1 AQL 0.065–0.65%



Food & Agriculture (SASO, SFDA)

Imported dates, wheat, rice: microbial & pest sampling. AQL 2.5–6.5% by SASO 4850 / Codex Alimentarius



Construction (NEOM, PIF projects)

Sampling of rebar, cement, tiles from Chinese/Indian suppliers — ASTM/ISO acceptance sampling



Electronics (ZATCA, SMC)

Electronic components for military/industrial use — MIL-STD-1916 zero-acceptance sampling (c=0)

Acceptance Sampling vs. Alternatives

Choosing the right quality tool for the right context

Factor	100% Inspection	Acceptance Sampling	SPC / Process Control
Cost	Very High	Moderate	Low (once established)
Speed	Very Slow	Fast	Real-time
Applicable when...	High-value/critical items	Large lots from supplier	Ongoing production process
Requires...	All items accessible	Random sample + lot integrity	In-process data & control charts
Error type	Human error risk (fatigue)	Sampling risk (α & β)	Type I/II in control charts
Best for	Aerospace life-critical parts	Incoming raw materials	Manufacturing line monitoring
Example	Heart implant device testing	SFDA drug lot inspection	Bottling line fill weight control

Common Pitfalls & Best Practices

Lessons from industry practitioners

Common Pitfalls

- Non-random sampling — pulling only top-layer items from a pallet
- Using wrong AQL for the risk level (too lenient for critical parts)
- Ignoring lot size — same n for 100 vs. 10,000 units gives very different protection
- Not updating the plan when supplier quality improves or deteriorates
- Treating sampling as a replacement for process improvement
- Accepting a 'failed' lot informally due to delivery pressure

Best Practices

- Always use truly random sampling (systematic or stratified random)
- Set AQL based on product criticality: 0.065% for life-critical, 4% for cosmetic
- Document the sampling plan in the supplier contract (PO or QAA)
- Switch to tightened inspection after 2 consecutive rejections
- Use reduced inspection for consistently high-performing suppliers
- Record all results — they form the basis for supplier scorecards



Concept Check

Quick questions to test your understanding

1. A company uses $n=100$, $c=3$. They find 4 defectives. What happens?

A) Accept the lot

B) Reject the lot ✓

C) Sample again

D) Accept with warning

2. AQL = 1% and $\alpha = 0.05$ means:

A) 1% of lots are defective

B) 95% of lots are rejected

C) 5% chance of rejecting a good lot ✓

D) 1% consumer risk

3. Which curve shows better discrimination between good and bad lots?

A) Flatter OC curve

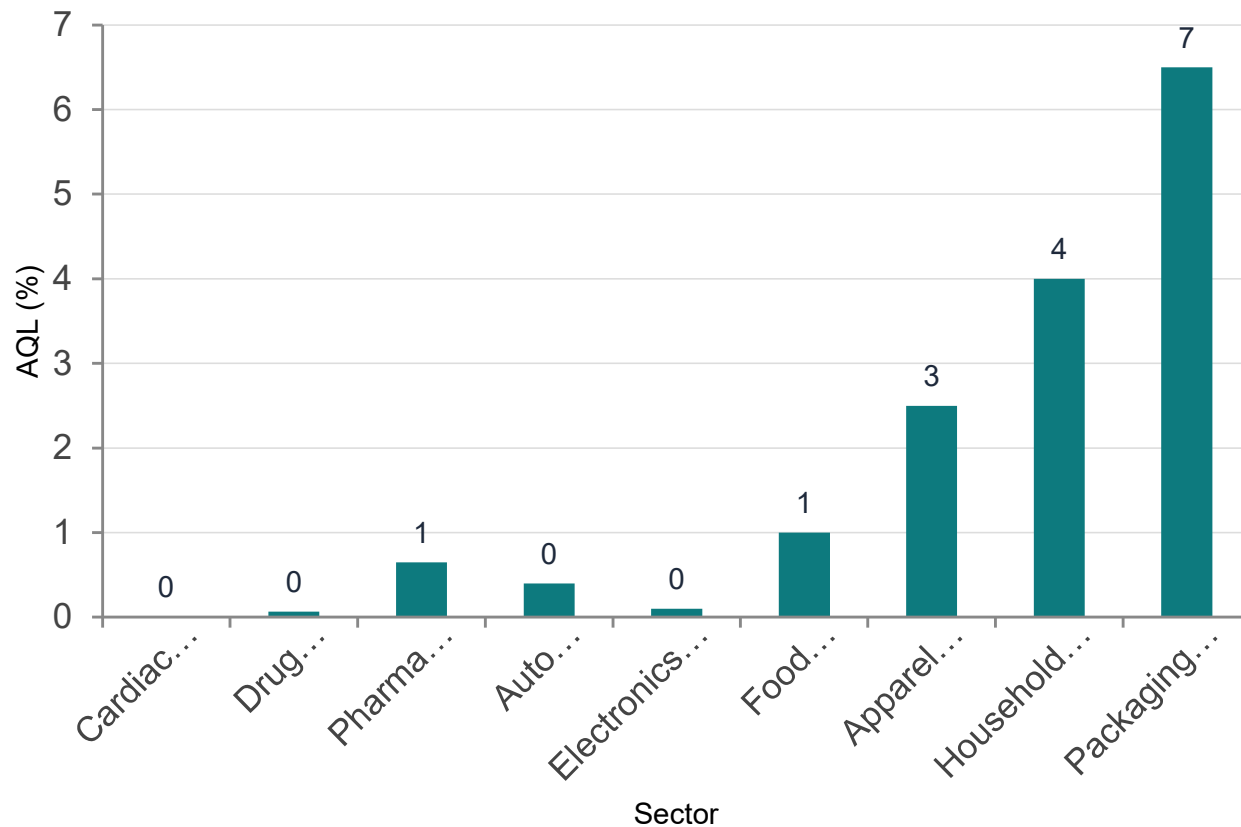
B) Steeper OC curve ✓

C) Both are equal

D) Depends on AQL only

Choosing the Right AQL: Industry Guide

AQL varies dramatically by risk level and sector



Source: ISO 2859-1 Annex B | SFDA Guidelines | FDA 21 CFR Part 820 | ASQ Body of Knowledge

Risk Scale

AQL < 0.1%

Life-critical
Zero tolerance

AQL 0.1–1%

High risk
(safety, pharma)

AQL 1–2.5%

Medium risk
(functional)

AQL 2.5–6.5%

Low risk
(cosmetic, packaging)

Exit Ticket & Conclusion Actions

Before you leave today...



EXIT TICKET (3 minutes — write on paper or phone notes)

1

Name ONE industry in Saudi Arabia that uses acceptance sampling and explain WHY they use it.

2

In your own words: what is the difference between producer's risk (α) and consumer's risk (β)?

3

If $n=50$ and $c=2$, and you find 3 defectives — what is your decision, and what error might you be making?



Conclusion Actions

Read: Foster Ch.9A + ISO 2859-1 summary **Practice:** OC curve Excel exercise (posted on Blackboard) **Explore:** SFDA sampling guidelines at sfda.gov.sa

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